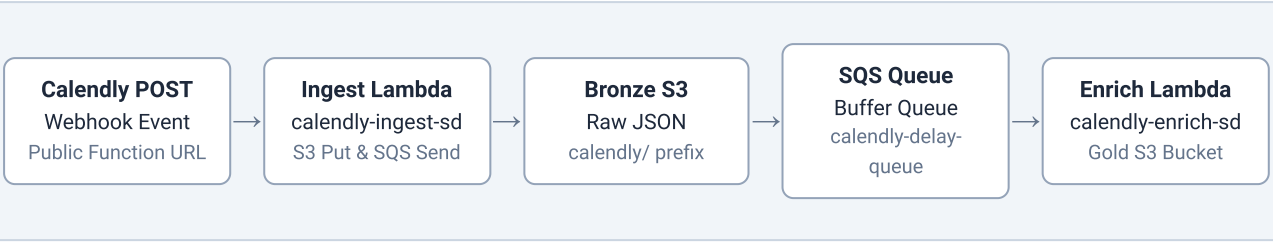
Calendly Webhook Pipeline Guide

AWS Serverless Real-Time Medallion Architecture (Bronze to Gold) Implementation

AWS REGION	ARCHITECTURE	INGESTION AUTH	ORCHESTRATION
us-east-2 (Ohio)	Medallion (Bronze/Gold)	Function URL (NONE)	Event-Driven SQS Trigger

1. Real-Time Webhook Architecture

The Calendly pipeline ingests real-time meeting scheduling webhooks, stores raw JSON payloads in a Bronze S3 bucket, buffers processing messages through SQS, and enriches invitee and event details into structured Gold JSON records.



2. Step 1: Queue & Endpoint Setup

- Create SQS Standard Queue: `calendly-delay-queue` in `us-east-2`.
- Create Lambda: `calendly-ingest-lambda-sd` with Environment Variables: `BRONZE_BUCKET_NAME` (`insightflow-bronze-bucket-sd`) and `SQS_QUEUE_URL`.
- Enable **Function URL** with Auth Type `NONE` and resource-based policy granting `lambda:InvokeFunctionUrl` and `lambda:InvokeFunction` to principal `*`.

3. Step 2: Deploy Ingestion Lambda Code

```
import json, os, boto3 s3_client = boto3.client('s3') sqs_client = boto3.client('sqs') BRONZE_BUCKET = os.environ.get('BRONZE_BUCKET_NAME', 'insightflow-bronze-bucket-sd') SQS_QUEUE_URL = os.environ.get('SQS_QUEUE_URL') def lambda_handler(event, context): try: body = event.get('body', json.dumps(event)) payload = json.loads(body) if isinstance(body, str) else body event_uuid = payload.get('payload', {}).get('event', {}).get('uuid', 'unknown_event') s3_key = f"calendly/calendly_event_{event_uuid}.json" s3_client.put_object( Bucket=BRONZE_BUCKET, Key=s3_key, Body=json.dumps(payload), ContentType='application/json' ) if SQS_QUEUE_URL: message_body = {"event_uuid": event_uuid, "s3_key": s3_key, "bucket": BRONZE_BUCKET} sqs_client.send_message(QueueUrl=SQS_QUEUE_URL, MessageBody=json.dumps(message_body)) return {'statusCode': 200, 'body': json.dumps('Calendly webhook received and queued successfully!')} except Exception as e: return {'statusCode': 500, 'body': json.dumps(f"Error: {str(e)}")}
```

4. Step 3: Configure & Deploy Enrichment Lambda

Function Name: `calendly-enrich-lambda-sd` | Trigger: **SQS** (`calendly-delay-queue`)

```
      import json, os, boto3 s3_client = boto3.client('s3') GOLD_BUCKET = os.environ.get('GOLD_BUC
KET_NAME', 'insightflow-gold-bucket-sd') def lambda_handler(event, context): records = event.get
('Records', [{"body": json.dumps(event)}]) for record in records: body = json.loads(record.get('bo
dy', json.dumps(event))) payload = body if isinstance(body, dict) else json.loads(body) s3_key = p
ayload.get('s3_key') bucket = payload.get('bucket') if s3_key and bucket: obj = s3_client.get_obje
ct(Bucket=bucket, Key=s3_key) calendly_data = json.loads(obj['Body'].read().decode('utf-8')) else:
calendly_data = payload event_obj = calendly_data.get('payload', {}).get('event', {}) invitee_obj
= calendly_data.get('payload', {}).get('invitee', {}) event_uuid = payload.get('event_uuid') or ev
ent_obj.get('uuid', 'unknown_uuid') gold_record = { "event_uuid": event_uuid, "event_name": event_
obj.get('name', 'Unknown Event'), "invitee_email": invitee_obj.get('email', 'unknown_email'), "inv
itee_name": invitee_obj.get('name', 'Unknown Name'), "status": invitee_obj.get('status', 'activ
e'), "start_time": event_obj.get('start_time'), "end_time": event_obj.get('end_time'), "raw_payloa
d": calendly_data } s3_client.put_object( Bucket=GOLD_BUCKET, Key=f"gold-calendly-{event_uuid}.jso
n", Body=json.dumps(gold_record), ContentType='application/json' ) print(f"Successfully wrote Gold
Calendly record for event {event_uuid}") return {'statusCode': 200, 'body': json.dumps('Calendly e
nrichment completed!')}
```

5. Postman End-to-End Test Verification

> **Validation Confirmation**
> A mock `invitee.created` webhook posted via Postman successfully landed in Bronze (`calendly_event_test-event-uuid-1234.json`) and processed into Gold (`gold-calendly-test-event-uuid-1234.json`).

6. Calendly Gold Schema Specification

Raw Webhook Path	Gold Schema Field	Data Type	Description
`payload.event.uuid`	`event_uuid`	String	Unique Calendly meeting event ID
`payload.event.name`	`event_name`	String	Title of scheduled meeting / event type
`payload.invitee.email`	`invitee_email`	String	Email address of the booking invitee
`payload.invitee.name`	`invitee_name`	String	Full name of the booking invitee
`payload.invitee.status`	`status`	String	Booking status (e.g., `active`, `canceled`)
`payload.event.start_time`	`start_time`	ISO8601 Timestamp	Scheduled start timestamp (UTC)
`payload.event.end_time`	`end_time`	ISO8601 Timestamp	Scheduled end timestamp (UTC)